

(4.2)

Naše najbl polynom najvije čvorov stupni  $p(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$ , koga slijedi  $p(0) = 1$ ;  $p(1) = 2$ ;  $p(2) = -3$

a po sebi je  $\sum_{k=0}^4 a_k^2$  minimalni.

U zadatku pje, da  $p(0) = 1 = a_0$

$$p(1) = 2 \Rightarrow a_4 + a_3 + a_2 + a_1 = 1$$

$$p(2) = -3 \Rightarrow 16a_4 + 8a_3 + 4a_2 + 2a_1 = -4$$

Skraćivši matricu  $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 8 & 4 & 2 & 1 \end{pmatrix}$  u vektoru pravi vektor  $\vec{b} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ . Specijalne OG po  $\vec{b}$  na  $\text{Im } A^T = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 8 \\ 4 \\ 2 \\ 1 \end{pmatrix} \right\}$

Poskušajmo tek, jeli bilo kakav na prethodne 11.3.:

1) Najdimo  $\vec{z}$  k. z.  $\underbrace{AA^T \vec{z}}_{\vec{b}} = \vec{b}$

$$AA^T = \begin{pmatrix} 4 & 45 \\ 45 & 85 \end{pmatrix} \quad \begin{pmatrix} 4 & 45 & 1 \\ 45 & 85 & -2 \end{pmatrix} \rightarrow \vec{z} = \begin{pmatrix} 1 \\ -1/5 \end{pmatrix}$$

$$2) \text{ Najdimo blizak vektor } \vec{z} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} : \vec{z} = A^T \vec{z} = \begin{pmatrix} -3/5 \\ 1/5 \\ 3/5 \\ 1/5 \end{pmatrix}$$

Blizak polynom koji je:  $p(x) = -\frac{3}{5}x^4 + \frac{1}{5}x^3 + \frac{3}{5}x^2 + \frac{1}{5}x + 1$